NUTTHAWAT PANYANGNOI (209) 407-9730
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EDUCATION

Bachelor of Science: **Computer Engineering (GPA: 3.4/4.0)** **Graduated 2025**
San Jose State University (San Jose, CA)

Associate of Science: **Computer Science and Minor in Mathematics (GPA: 3.4/4.0)** **Graduated 2022**
Las Positas College (Livermore, CA)

Relevant Coursework: Programming Concepts and Methodology C/C++, Object-Oriented Program C++, Algorithms and Data Structure Design C/C++, Compiler Design, SQL Database Systems, Operating Systems, Computer Networks. Machine Learning for Big data, Real-Time Embedded Co-Design, Digital Design, Microprocessor Design and Computer Architecture and Design.

CERTIFICATIONS

NASA L'SPACE Mission Concept Academy Certificate (Research on NASA's discovery mission project management) 1/2024
NASA Proposal Writing and Evaluation Experience Certificate (Research on high-voltage Lithium-ion battery) 12/2024

WORK EXPERIENCE

Hardware Test validation (Platforms infrastructure team) (Extended Workforce at Google) **10/2025 to Currently**
Tezerakt LLC **Mountain View, CA**

- Execute extreme hot and cold thermal cycling on unreleased Google TPUs to validate hardware reliability.
- Developed automation scripts (shell, python) to monitor on DUTs status, network uplink and hardware validation
- Performed Google Proprietary OS deployment via Net booting and managed hardware remote using BMC

IT Technician & specialist (Full time) **03/2020 to 08/2023**
Mountain Cascade, Inc. **Livermore, CA**

- Identifying, troubleshooting, and resolving hardware and software issues in corporate environment.
- Performed hands-on technical work such as hardware repair, system upgrade, BIOS flashing, RAID configuration, network configuration, VPN network setup, and regular maintenance of servers and office workstations.

Frontend Software Developer (Full time) **04/2018 to 03/2020**
Amber Systems Technology **Pleasanton, CA**

- Designed and refined the user interface of POS systems to ensure smooth navigation, visual appeal, and ease of use for small business clients and developed full-stack web applications provided long-term scalable solutions.
- Led front-end development of custom web application software by collaborating with clients to understand their goals and turning those into functional features.

PROJECT EXPERIENCES

Handheld Computing Device (C, ASM, Linux, KiCad, Raspberry Pi CM4) **In Progress**

- Designing and developing a handheld computer powered by Raspberry Pi CM4 on a custom Clockwork mainboard.
- Built a lightweight Linux Operating System with Yoctor, C program to toggle with MCU, L2C, GPIO, etc.

Neural Network Classifier – Iris Dataset (PyTorch, NumPy, Pandas, Python, Scikit-learn) **May 2025**

- Built a supervised neural network model to classify Iris flower species, leveraging linear regression as a baseline for comparison.
- Implemented model training with Adam optimizer and cross-entropy loss, tracking evaluation accuracy on test data.

Miscellaneous Embedded Projects in classroom (RTOS, C) **Apr 2025**

- Developed a C program to toggle on the Texas Instrument MCU MSP432 in RTOS, internal, external oscillators
- Implemented peripheral function to interact with the MCU such as AES-128 encryption, access Flash memory, SRAM, send text through UART port, create ADC digitize Timer Trigger, measure ADC temperature, etc.

FPGA implement RISC-V CPU Pipeline (Vivado, Verilog, System Verilog) **Apr 2025**

- Developed and design CPU pipeline for RISC-V architecture using Verilog and AI tools (for learning purpose)
- Implemented circuit using Altera modalism simulated mux's, ALUs, data registers, Finite state machines, CPU design, etc.

Square & Triangular Waveform Generator Circuit (LTspice, Schmitt-Trigger CD-40106, Oscilloscope) **May 2024**

- Designed and simulated a waveform generator using LTspice to produce square and triangular wave outputs across varying frequencies, leveraging a Schmitt-Trigger (CD-40106) and BC547B transistor.
- constructed and tested the circuit on a breadboard to verify functionalities with an oscilloscope.
- Applied circuit analysis skills such as RLC circuit, op amps, MOSFET, diodes, transistors, Oscilloscope, etc.

Note Hub Web Application (Python3, Flask, SQL Alchemy, Django) **Apr 2023**

- Built a web-based note-taking application inspired by Notion, allowing users to create, update, and delete notes.
- Implemented secure login system and session management using Flask and SQLite for lightweight deployment.

EXTRACURRICULAR ACTIVITIES

Las Positas College Computer Engineering Club **Active member - 2022**

- Weekly collaborated on C++ projects leveraging data structure to implement efficient algorithm solved complex problems.

San Jose State Machine Learning Club **Active member - 2025**

- Monthly collaborated on projects applying machine learning techniques (Neural Network, NLP models, etc.)
- Participated on latest ML technology and workshops and technical discussions outside the classroom.

San Jose State Cube3 Satellite Club (We built Satellite) **Active member - 2025**

- Developed embedded C programs for the Astraeus-1 control board, enabling communication across multiple microcontrollers and integrating onboard sensors (XBee PRO S38 radio, barometer Mel3115, motion tracker ICM 20948, GPS, NEO M9N rope meter level GNSS position, and Micro Mod MCU).
- Partially implemented real-time data processing pipelines and built a React web interface to display system performance metrics and sensor outputs.

San Jose State Honor Engineering Club Tau Beta Pi **Active member - 2025**

- Recognized for academic excellence and active participation in Tau Beta Pi, engaging in professional development events such as guest lectures and engineering seminars.

San Jose State Software Engineering Club **Active member - 2025**

- Participated in coding challenges, technical talks, and peer collaboration events.